SARTHAK SHRIVASTAVA

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EDUCATION

Indian Institute of Technology Delhi, India *2019 - 2023*
Bachelor of Technology (B.Tech.), Civil Engineering *GPA: 8.283/10.0*

AREAS OF INTEREST

Computational Structural Mechanics, Continuum Mechanics, Finite Element Methods, Multi-scale and Multi-physics Modeling, Fracture Mechanics, Peridynamics, Molecular Dynamics, Structural Dynamics, Structural Health Monitoring, Numerical Methods, Machine Learning

RESEARCH EXPERIENCE

Research Assistant | *Multiphysics & Multiscale Mechanics Research Group (M3RG), IIT Delhi* (*Jan, 2023 - Present*)

- Conducting numerical simulations using finite element and peridynamics methods to investigate the role of indenter geometry on indentation-induced damage of glasses under the supervision of Prof. N. M. Anoop Krishnan

Summer Undergraduate Research Fellow | *Industrial Research & Development Unit, IIT Delhi* (*May, 2021 - Oct, 2021*)

- Worked on a summer research project titled 'The Application of Machine Learning to Structural Health Monitoring' under the supervision of Prof. Sahil Bansal, funded by the Industrial Research & Development (IRD) Unit, IIT Delhi

PROJECTS

Peridynamics-Based Modeling of Nanoindentation Response of Silicate Glasses (*Sep, 2022 - Present*)
Advisor: Prof. N. M. Anoop Krishnan, IIT Delhi

- Developing a **peridynamics**-based numerical model to investigate the nanoindentation behavior of silicate glass under **varying tip angles of the indenter**, providing insights into the nanoscale mechanical behavior of the material
- Validating the numerical model using **finite element analysis** (FEA), demonstrating the accuracy and reliability of the proposed model in predicting the **nanoindentation response** of silicate glasses
- Conducting **upscaling analysis** to investigate the macroscopic behavior of silicate glass under different indentation conditions, providing a better understanding of the mechanical properties of the material at the macroscale level

Stability of Steel Columns at Elevated Temperatures | *Finite Element Analysis* (*Nov, 2022 - Dec, 2022*)
Advisor: Prof. Vasant Matsagar, IIT Delhi

- Eigen value problem to determine the **buckling load** through FE formulation showed convergence with Euler's formula
- Stresses were evaluated under **thermo-mechanical loading** for various boundary conditions of the column
- Performed **non-linear analysis** of columns using **Riks analysis**; determined **critical thermo-mechanical stresses** and facilitated the performance of **post-buckling analysis** until collapse

The Application of Machine Learning to Structural Health Monitoring | *UROP* (*May, 2021 - Oct, 2021*)
Advisor: Prof. Sahil Bansal, IIT Delhi

- Devised a **vibration-based damage-detection** approach using **AI methods** on time response simulated data
- Extracted different **damage-sensitive features** based on **signal statistics** and **modal properties**; analyzed **signal response time-series** using **AR and ARX models**; proposed **statistical model for feature discrimination**
- Ranked damage-sensitive features and employed **support vector machines** to classify **structural condition states**

Transverse Vibration Analysis of Euler-Bernoulli Beam | *Computational Mechanics* (*Mar, 2022 - Apr, 2022*)
Advisor: Prof. Prapanch Nair, IIT Delhi

- Derived an **explicit finite difference scheme** with **second order accuracy** by discretizing the governing equations
- Implemented Python script for computation of **natural frequencies** of vibration, **mode shapes**, and **vibration response** of the elastic beam for **different boundary conditions**, viz. fixed-fixed, fixed-pinned, pinned-pinned, and fixed-free
- **von Neumann analysis** was done to establish **stability requirements** of the solution, explicit FD scheme was found to be **conditionally stable** and **convergent**; compared the results with analytical solution to validate the approach

Structural Design using STAAD.Pro | *Structural Design & Detailing* (*Sep, 2022 - Nov, 2022*)

- Designed and analyzed the structures of **hostel campus** comprising five blocks of **G+10 storey buildings**, **reinforced concrete overhead water tank**, **industrial shed truss**, and **transmission tower** using AutoCAD and STAAD.Pro
- Resistance to **earthquake** and **wind loads** were incorporated into the structural design process as per Indian Standards

Food Outlet Management System | *Data Structures & Algorithms* (*Apr, 2021*)

- Developed a simulation environment for a food outlet having facility of **multiple ordering/billing counters** and food delivery on **first-order-first-serve basis**; implemented **Priority Queue**, **AVL Tree**, **Array Deque** based data structures
- Statistics like **average waiting time**, **average queue length** were computed in order to **improve customer service**

TEACHING & MENTORING EXPERIENCE

Tutor | *Course Hero, Inc.* (*Dec, 2021 - Apr, 2023*)
- Offered online tutoring to students seeking academic assistance in Physics, Mathematics, and Civil Engineering
- Received **consistently positive ratings** and feedback from students for effectiveness and clarity of instruction

Academic Mentor | *Board of Student Welfare (BSW), IIT Delhi* (*Nov, 2020 - Feb, 2021*)
- Provided **peer mentorship** and tutoring to undergraduate freshers in **Engineering Mechanics** (APL100) course
- Assisted the students to acclimate to a new academic setting by **facilitating weekly study sessions**

Educator (IIT-JEE Physics) | *Unacademy* (*Aug, 2020 - Oct, 2020*)
- Taught Advanced Physics to **500+ IIT-JEE aspirants**; held interactive sessions and emphasized **analytical thinking**, **problem-solving** and **exam-taking** skills through the use of **visual learning** approaches; offered learning resources
- Ranked among **top 100 educators** in IIT-JEE category with over **15K minutes** of watch time on online platforms

Mentor | *Project Aarohan, National Service Scheme (NSS), IIT Delhi* (*Sep, 2019 - Apr, 2020*)
- Mentored underprivileged students of class XI and XII for competitive exams
- Taught Physics; provided them with comprehensive study material and conducted doubt clearing sessions

WORK EXPERIENCE

Summer Analyst | *Axis Bank (Business Intelligence Unit)* (*Jun, 2022 - Jul, 2022*)
- Built a **web-scraping tool** that extracts information related to employees' salaries, revenues, capitals of an organization
- Employed **Selenium** to **automate web browser** and interact with **dynamic web pages**; handled **captcha recognition** through **OpenCV** based **character segmentation** technique; analyzed the scraped data to gain useful **business insights**
- Developed tool is capable of scraping data of **600K+ organizations**; proposed ideas to **amplify sales** of salary accounts

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, Julia, MATLAB
- **Software & Tools:** Microsoft Office, High Performance Computing, STAAD.Pro, Abaqus FEA, Ansys, LaTeX, Git
- **Libraries:** C++ STL, Pandas, NumPy, SciPy, Matplotlib, Scikit-learn, Peridigm, Selenium, Tesseract-OCR, OpenCV

SCHOLASTIC ACHIEVEMENTS

- Awarded the **Summer Undergraduate Research Award (SURA), 2021** from Industrial R&D Unit, IIT Delhi *2022*
- **Online Judges:** Scored **5-star rating** on ***HackerRank*** and **3-star (max. 1783 rating)** on ***CodeChef*** *2021*
- **JEE (Advanced):** Secured **All India Rank 1841 (top 1 percentile)** among 0.17 million candidates *2019*
- **JEE (Main):** Achieved **All India Rank 1403 (top 0.1 percentile)** among 1.2 million candidates *2019*

PRESENTATIONS

Impact Studies with Peridynamics, Civil Engineering Seminar, Indian Institute of Technology, Delhi, India. Apr 2022.

REFERENCES

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